March 15, 2025

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2415 Eisenhower Avenue
Alexandria, VA 22314
Via email: [redacted]

Subject: DCC Comments on the Development of an Artificial Intelligence (AI) Action Plan

To Whom It May Concern:

The Data Center Coalition (DCC) appreciates the opportunity to provide comments in response to the Request for Information (RFI) on the AI Action Plan issued by the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation (NSF) on behalf of the White House Office of Science and Technology Policy (OSTP) on February 6, 2025. DCC is the membership association for the U.S. data center industry, representing leading data center owners and operators, as well as companies that lease large amounts of data center capacity.[1] DCC's member companies provide the digital infrastructure that supports the applications, capabilities, and essential services that enable the modern economy, including cloud computing and artificial intelligence (AI). DCC member companies are making significant, multi-billion dollar investments in U.S. data center infrastructure. DCC empowers and champions the data center community through public policy advocacy, thought leadership, stakeholder outreach, and community engagement. The comments we are providing focus on the need to unleash the infrastructure and workforce necessary to power, supply, and operate the nation's data centers.[2]

I. Background

There is unprecedented demand for the digital services that have become central to our daily lives and modern economy. With an average of 21 connected devices per household in the U.S.,[3] the essential role of data centers is expected to grow as consumers and businesses generate twice

[1] The Data Center Coalition's website address is www.datacentercoalition.org. Public testimony and written comments submitted by DCC do not necessarily reflect the views of each individual DCC member.
[2] This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.
[3] Deloitte, "Connect consumer survey 2023," (September 2023).

as much data in the next five years as they did in the past decade.[4] This growth is driven by the widespread adoption of cloud services, the proliferation of connected devices, and the rapid scaling of advanced technologies like generative AI, which alone could create between $2.6 trillion and $4.4 trillion in economic value globally by 2030.[5]

The data center industry's total contribution to U.S. GDP was $3.5 trillion between 2017 and 2023.[6] Notably, the industry's growth has surpassed that of the overall U.S. economy in recent years. From 2017 to 2023, direct employment in the U.S. data center industry expanded by over 50 percent, compared to 10 percent growth in employment for the United States overall during the same timeframe.[7]

Data centers also enable critical and emerging technologies that safeguard national security and underpin America's global competitiveness. Leadership in developing and operating AI in the United States is vital for protecting national security and ensuring that AI systems are safe, secure, and trustworthy. As Vice President JD Vance has noted, "[T]his administration will ensure that American AI technology continues to be the gold standard worldwide and we are the partner of choice for others—foreign countries and certainly businesses—as they expand their own use of AI."[8]

Data centers are the backbone of AI and the digital economy, driving innovation and growth across various sectors. DCC members, including developers and operators of hyperscale, self-perform/enterprise, build-to-suit, multi-tenant/colocation, and edge data centers, are at the forefront of this transformation. While common considerations such as access to land, power, equipment, and a skilled workforce are crucial, there is no one-size-fits-all solution. By proactively removing barriers to deployment, the federal government can play a pivotal role in supporting this critical sector, fostering economic growth, and maintaining our competitive edge against foreign adversaries.

II. Policy Recommendations

DCC thanks the Trump administration for its support of the American AI industry with the Executive Order "Removing Barriers to American Leadership in Artificial Intelligence" (AI EO) on January 23, 2025 and the President's desire to make the United States an undisputed leader in the development and application of AI technology. As stated in the AI EO, "It is the policy of the United States to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security." DCC members embrace this

[4] JLL, "Data Centers 2024 Global Outlook," (2024).
[5] McKinsey, "The economic potential of generative AI: The next productivity frontier," (June 2023).
[6] PwC, "Economic Contributions of Data Centers in the United States," (February 2025).
[7] Ibid.
[8] Remarks by the Vice President at the Artificial Intelligence Action Summit in Paris, France (February 11, 2025).

opportunity to advance the President's agenda by building data center infrastructure and enabling AI development in the U.S.

Collaboration between the federal government and the private sector is essential to remedy regulatory barriers, loosen supply chain bottlenecks for the construction and operation of this critical infrastructure, and ensure that the U.S. is on pace with the Trump administration's vision for leading global development and deployment of AI.

Below we provide a comprehensive series of policy recommendations to be included in the AI Action Plan. Our recommendations encompass potential federal agency and congressional action and focus on the removal of barriers to faster data center construction and operation, ranging from power access to supply chains to workforce development.

1. Energy Infrastructure and Access to Power

The U.S. requires more data centers to provide the computing power needed to support critical and emerging technologies, including AI, that deliver broad public and economic benefits. As demand for data center services continues to rise, timely and reliable access to power is the pacing challenge for the industry. After nearly two decades of relatively flat electricity consumption, the U.S. is experiencing a significant increase in power demand driven by several economic growth trends, including the onshoring of new manufacturing, widespread electrification, hydrogen fuel production, and growth in demand for data center services. As President Trump noted, "the big problem is we need double the energy we currently have in the United States…for AI to really be as big as we want to have it because it's very competitive—it will be very competitive with China and others."[9]

Data center energy demand supports mission-critical operations—and businesses of all sizes—in a 24/7, always-on, digital environment, driving economic growth in our local communities and maintaining our leadership on the global stage. Data center investment catalyzes supply chain and service ecosystems, provides long term employment for the construction workforce, and provides high quality jobs to support ongoing operations. At the national level, each direct job in the data center industry supports more than six jobs elsewhere in the U.S. economy.[10]

DCC believes the AI Action Plan should focus on ensuring expedient access to reliable, timely, and affordable energy to secure U.S. leadership in the global digital infrastructure race. The federal government plays a key role in promoting policies that enable greater expansion of the data center industry. As Secretary of Energy Chris Wright noted in his confirmation hearing, "I am committed to growing our electricity grid and our energy production and removing those

[9] President Trump Delivers Virtual Remarks at World Economic Forum (January 23, 2025).
[10] PwC, "Economic Contributions of Data Centers in the United States," (February 2025).

barriers that are standing in the way."[11] We look forward to engaging with the White House and relevant agencies on the implementation of these recommendations and reforms. DCC believes that robust dialogue and coordination between the federal government, industry, and other stakeholders is necessary to ensure America wins the AI race.

a. *Ensuring timely access to reliable energy for all customers*

The data center industry's greatest challenge is timely access to reliable energy. Energy policy must evolve by promoting robust transmission planning, faster interconnection processes, innovative commercial arrangements, and next generation technologies. We believe the following policy recommendations should be included in the AI Action Plan:

<u>Administrative Action:</u>

- **Promote Collaboration Across the Energy Landscape:** Collaboration and cooperation from state and federal regulators, policymakers, grid planners, and others is needed to ensure timely access to energy for data centers. The federal government should play a leading role in ensuring that these various stakeholders work together to advance data center development.

- **Advance Onsite Cogeneration:** Also known as Combined Heat and Power (CHP), onsite cogeneration is an available dispatchable capacity that can serve as either baseload power or offset fluctuations of variable renewable energy resources. While the Department of Energy (DOE) invested in a CHP program in the early 2000s, its interest in the technology has waned in recent years. Advancing support for this technology at the federal level can improve both reliability and efficiency, and help address capacity issues at both data centers and other critical facilities.

- **Support Federal Lands for Data Center Development:** The federal government should consider the use of federal lands, both brownfield and greenfield, for data center development. The federal government should also explore how to bring a variety of power access solutions—from onsite behind-the-meter to local utility arrangements—into any federal land projects. This should also come with consideration of how the government can speed permitting timelines on federal lands.

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[11] Senate Energy and Natural Resources Committee Holds Hearing on the Nomination of Chris Wright (January 15, 2025).

Joint Congressional and Agency Action:

- **Federal Support for Utility Investment:** Increasingly, utilities are concerned about risk associated with building out generation and grid capacity required to serve large load customers, such as data centers. Some utilities are requiring substantial financial assurance requirements from new large load customers that tie up large amounts of capital and credit for extended periods of time, chilling investment in some markets and stranding capital that would otherwise support investment in additional data centers and markets. Congress and the administration should explore changing existing programs and mechanisms or establishing new ones that can provide utilities with confidence in building out required power generation and infrastructure while easing the chilling effects of these new and increasing financial requirements on large load customers. Existing programs that could facilitate loan guarantees, credit insurance, or revolving funds include the DOE's Loan Programs Office, the Department of Agriculture's (USDA) Rural Utilities Service, or the Department of Defense's Defense Community Infrastructure Program.

- **Support Next Generation Technologies:** Regulatory frameworks should account for innovation such as hydrogen fuel cells, advanced geothermal development, carbon capture and storage (CCS), energy storage, and other domestically sourced technologies that will undoubtedly be utilized if barriers to deployment are removed. Tax incentives, faster permitting, and national deployment programs can unleash these technologies at scale—not only for data centers, but also for other industries

Congressional Action:

- **Optimization of the Current Grid:** Congress could clarify how transmission owners need to deploy Alternative transmission technologies (ATTs) and Grid Enhancing Technologies (GETs) in Section 219(b)(3) of the Federal Power Act (FPA) and provide an incentive structure and regulatory clarity that will create cost-effective solutions for grid resilience, security, and reliability.[12] Congress can also direct funding to reconductor existing transmission lines, which could quadruple new transmission capacity by 2035.[13]

- **Support Nuclear Cost Overrun Insurance:** Nuclear power will be a key part of a suite of new energy infrastructure built to meet future energy demand. Congress should advance legislation, such as the Accelerating Reliable Capacity (ARC) Act introduced by Senator Risch, that aims to address the biggest hurdle in commercial nuclear power

[12] U.S. Congress, House, *Advancing Grid-Enhancing Technologies Act* (Advancing GETs Act), H.R.. 7624, 118th Cong., 2nd sess., introduced in House March 12, 2025.
[13] GridLab, "Reconductoring Technical Report," (April 2024).

deployment—financial risk.[14] The bill provides a limited risk reduction program for the first few emerging nuclear technology projects to demonstrate the technology's safety and reliability. Beyond risk mitigation, more federal funds will be needed to jumpstart deployment, given the significant expense of new nuclear projects.

Federal Energy Regulatory Commission (FERC) Action:

- **Ensuring Load Forecasting is Robust:** High confidence in load forecasting is necessary to spur investment in much-needed new generation resources and transmission infrastructure. FERC can help improve the transparency, standardization, and effectiveness of load forecasting processes. We respectfully recommend that FERC initiate an inquiry into best practices around load forecasting and/or establish FERC-led workshops that support the development of best practices for load forecasting over multiple time horizons across all regions in the U.S.[15]

- **Promote Proactive Transmission Planning:** The federal government should support proactive strategies to prepare for expected load increases in the transmission planning process. FERC should ensure robust and timely implementation of Order 1920-A, which requires transmission providers to conduct long-term planning for regional transmission facilities over a 20-year time horizon to anticipate future needs and to determine how to pay for those transmission facilities.[16]

- **Explore Generation Interconnection Reforms:** FERC should continue to evaluate options to expedite generation interconnection requests. Recent reports suggest generator interconnection timelines can take up to four years in some regions of the country.[17] Grid operators and FERC should continue to explore ways to use surplus interconnection rights to better utilize and quickly connect generation units.[18] Additionally, FERC should instruct RTOs/ISOs to integrate AI and automation into the interconnection process to speed up lengthy study timelines. The Midcontinent Independent System Operator (MISO) has seen promising benchmarking results showing that automating the study process improves the timeline from 686 days to 10 days.[19]

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[14] U.S. Congress, Senate, *Accelerating Reliable Capacity Act* (ARC Act), S. 5421, 118th Cong., 2nd sess., introduced in Senate December 4, 2024.
[15] Data Center Coalition, Post-Technical Conference Comments in Docket No. AD24-11-000 Before the Federal Energy Regulatory Commission
[16] *Building for the Future Through Elec. Reg'l Transmission Planning & Cost Allocation*, Order No. 1920, 89 FR 49280 (June 11, 2024), 187 FERC ¶ 61,068 (2024).
[17] American Clean Power, "Interconnection 101," (June 2023).
[18] Utility Dive, "PJM's proposal to unlock unused interconnection capacity gains broad support," (January 2025).
[19] MISO Interconnection Process Working Group, "SUGAR Implementation," (March 4, 2025).

- **Require Interregional Transmission:** Interregional transmission is critical for ensuring grid reliability, reducing costs, and strengthening resilience. FERC should act to require transmission planning regions to proactively engage in interregional transmission planning, including establishing regionally specific minimum transfer capability guidelines.[20]

- **Access Priority for Certain Interconnection Services:** In Order 2003, FERC established two types of interconnection service: energy resource interconnection service (ERIS) and network resource interconnection service (NRIS), the latter of which is commonly thought to enable the generator to qualify as a capacity resource. However, the current study of ERIS resources does not meaningfully differ from NRIS, despite the potential efficiencies and faster interconnection times that ERIS can provide if properly defined and studied. The failure to make ERIS a viable option is exacerbating the interconnection challenge, unnecessarily raising costs, and compromising reliability through the inability to add new resources in a timely fashion. FERC should (1) update the definitions and clearly differentiate between ERIS, provisional ERIS, and NRIS interconnection standards, and (2) require a transparent, economics-informed screening step as part of the interconnection process to help inform developer decisions.

- **Ensure Co-Location Arrangements Remain an Option:** Co-location is one of several commercial arrangements that enable data centers to meet their growing power needs in an efficient manner, particularly in regions facing transmission or interconnection constraints.[21] President Trump has previously signaled strong support for co-location arrangements,[22] and as such the Trump administration should ensure FERC adopts a co-location policy that will dictate long-term market viability of co-location arrangements while ensuring fair treatment of all ratepayers.[23] Current uncertainty around regulatory expectations is hindering necessary industry investments; clear guidance is essential to support continued economic growth and infrastructure development.

- **Optimization of the Current Grid:** ATTs/GETs offer a cost-effective, near-term solution to transmission constraints. They require far less time—a matter of months—to install compared to traditional transmission construction timelines.[24] Section 219(b)(3) of the FPA directs FERC to promote investment in electric transmission infrastructure that

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[20] Joint Comments of Large Consumers, Interregional Transfer Capability Study in Docket No. AD25-4-000 Before the Federal Energy Regulatory Commission.
[21] Data Center Coalition, Post-Technical Conference Comments in Docket No. AD24-11-000 Before the Federal Energy Regulatory Commission.
[22] President Trump Delivers Virtual Remarks at World Economic Forum (January 23, 2025).
[23] Data Center Coalition, Post-Technical Conference Comments in Docket No. AD24-11-000 Before the Federal Energy Regulatory Commission.
[24] WATT Coalition, "Unlocking the Grid: Key Benefits of Grid Enhancing Technologies."

uses advanced technologies. Also, FERC should proceed with setting a requirement that transmission owners deploy ATTs/GETs in certain contexts on existing transmission lines while implementing an incentive structure to give utilities an opportunity for an adequate financial return if the ATTs/GETs replace the need for new or upgraded transmission lines.[25] [26]

b. The need for comprehensive permitting reform

We thank the Trump administration for the support of broad permitting reform to accelerate American economic growth, national security, and economic competitiveness. Transmission and generation constraints across the country are restricting economic growth, including data center development, and part of that constraint comes from permitting delays. The federal government has a critical leadership role in ensuring the nation has sufficient energy capacity needed to power America's growing economy, including additional transmission lines, energy generation, and the deployment of new technologies. DCC supports congressional and administrative action on comprehensive permitting reform. We believe the following policy recommendations should be included in the AI Action Plan:

<u>Congressional Action:</u>

- **NEPA Overhaul:** The original National Environmental Policy Act (NEPA) statute provided environmental review for instances where other laws and statutes did not already apply. Permitting reform should deploy NEPA not as an addition to federal permitting programs and other agency environmental reviews, but in situations where other statutes/agencies have no role.
    - *Immediate approvals:* For categories of projects where environmental impacts are well understood, either due to the nature or location of the project, Congress should establish approval criteria that enable project clearance without delay.
    - *Accelerated approvals:* For projects that may cause unique or significant negative local environmental impacts, Congress should establish a bifurcated process that documents the categories of projects at the outset and then focuses environmental review and permitting on any uniquely local conditions of a project on an accelerated timeline.
    - *Accelerated adjudications:* Once a project is approved, any adjudications for projects should include a final decision timeline of well under one year to ensure that protracted litigation does not undermine project viability.

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[25] *Implementation of Dynamic Line Ratings*, 187 FERC ¶ 61,201 (2024).
[26] WATT Coalition, "Frequently Asked Questions."

- *State and local conformity:* Eligibility for any federal infrastructure funding, tax incentives, or grants shall be conditioned on a state or locality conforming to the same framework and timeline for fast approval and adjudication of projects.

- **Limit Judicial Review:** Reduce the statute of limitations deadline from six years to 150 days (five months) for filing lawsuits against an agency action approving or denying the permitting of an energy project. This change will help reduce uncertainty, so project developers know if they have final approval to proceed. The risk to developers spending hundreds of millions of dollars on a project that could be halted due to regulatory and legal challenges raises the cost of capital and, ultimately, costs for consumers.

- **Expand FERC Authority to Permit Transmission Projects in the National Interest:** Expand the backstop authority of FERC to permit transmission projects deemed "in the national interest."[27] This would allow transmission developers to appeal to FERC for siting approval in instances where states are unwilling or unable to grant construction permits in a timely manner. This change can save between two and five years in the approval process.

- **Interregional Transmission Planning:** Require neighboring transmission planning regions to create joint plans for constructing interregional transmission with a common set of assumptions and time horizons, prioritizing the use of existing infrastructure and rights-of-way.[28]

- **Create Consistent Interregional Cost Allocation Metrics:** Create a standard definition of "transmission benefits" so that the costs of a transmission project can be allocated among the customers receiving service in a consistent, fair, and reasonable manner. Transmission line costs are already allocated according to the well-established principle that customers should pay in proportion to how much they benefit from the line; customers that do not benefit, do not pay costs associated with a given project.[29] These lines are only built if the benefits exceed the costs by a reasonable margin. The complication is that different regions use different formulas for determining cost allocation based on different definitions of "transmission benefits." Congress will need to direct agencies who can then develop the detailed benefits nomenclature.

- **Expand Categorical Exclusions for Certain Transmission Activities:** Direct the U.S. Department of the Interior (DOI) and the U.S. Department of Agriculture (USDA) to

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[27] U.S. Congress, Senate, *Energy Permitting Reform Act of 2024 (EPRA)*, S. 4753, 118th Cong., 2nd sess., introduced in Senate July 23, 2024.
[28] Ibid.
[29] Ibid.

create new categorical exclusions for the following activities related to transmission: building transmission facilities within rights-of-way corridors; upgrading existing transmission and grid infrastructure within rights-of-ways or on previously disturbed land; and deploying energy storage technologies on previously disturbed lands.[30] These categorical exclusions currently exist at DOE but not at DOI or USDA, which are more often responsible for reviewing projects in need of the exclusions.

Federal Agency Action:

- **Clean Water Act Section 404 Nationwide Data Center Uses Permit:** Data center development generally falls under Nationwide Permit 39 (Commercial and Industrial Developments), which (1) limits use beyond 0.5 acre (after which a more extensive individual permit is required) and (2) requires a Pre-Construction Notification (PCN) for any impact. The lengthy timelines for the approvals are not consistent with other national permits that have higher or no limits or have a threshold where a PCN is not needed, which allows immediate action. We recommend that the U.S. Army Corps of Engineers (USACE) develops a Nationwide Permit for Data Centers with robust notification and coverage thresholds.

- **Clean Water Act Section 404 Streamlining:** Additional permitting efficiency improvements at USACE should include:
    - Direct the Corps Regulatory Branches to limit jurisdiction to areas below ordinary high-water mark and immediately adjacent areas/uplands. "Area of Potential Effect" for Section 106 consultations should not extend beyond USACE jurisdiction.
    - Set a regulatory time limit for responding to jurisdictional delineations.
    - Provide a single contact within the Corps for large data center investments.
    - Provide a mechanism for applicants to reimburse staff time/department consultants for expedited permit reviews.
    - Issue a new "Memorandum to the Field" reaffirming that prior converted cropland is not considered a Water of the United States, and rescinding direction related to "change in use" kick-out (included in a 2005 Memorandum to the Field). This would be consistent with the pre-2015 regulatory definition of Waters of the United States.

- **Clean Air Act Rules for Stationary Engines:** Diesel generators provide critical backup power for data centers. However, EPA stationary engine rules have been under a court remand since 2016 regarding the allowed maintenance and testing ("non-emergency

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[30] Ibid.

operation") provisions for emergency generators, creating operational uncertainty and risk. EPA issued a request for comments in 2023.

- ○ Require EPA to respond to the June 2023 public comment period on stationary engines and retain the 50 hours of non-emergency operation for stationary engines classified as "emergency" to maintain data center operational flexibility.
- ○ EPA should issue guidance to regional offices to ensure renewable diesel conforming to ASTM D975 is an approved fuel in the RICE rules.

<u>Joint Congressional and Agency Action:</u>

- **Streamline Nuclear Permitting:** The Nuclear Regulatory Commission (NRC) should further streamline permitting to allow for more projects to qualify under the faster Environmental Assessment (EA), and follow the direction of the Fiscal Responsibility Act 2023 to complete EA assessments under 12 months. President Trump has already promoted the "technology inclusive" approach in his first term that enables reactors of all types to follow the same type of safety rules, including for new Small Modular Reactors (SMRs) and other new nuclear technologies. As nuclear technologies evolve, outdated restrictions related to siting and other planning considerations should ease, allowing for NRC-regulated technologies to be deployed in the locations where they are needed most. We support swift promulgation of NRC rules to advance the administration's agenda of removing hurdles to the deployment of cutting-age nuclear technologies.

Looking ahead, the challenges of load growth and resource adequacy will require a concerted effort among the federal government, utilities, customers, states, and other stakeholders. Data centers are key players in the ongoing transformation of the electricity sector, and our industry is eager to contribute to solutions that ensure reliable, affordable, and adequate power for all consumers. DCC is actively engaged with stakeholders nationwide and we support greater communication, collaboration, and transparency in planning for future load growth. DCC members are committed to paying their full cost-of-service for generation, distribution, and transmission.

Finally, even in places where co-location and other off-grid power arrangements may make sense, new transmission—both regional and interregional—as well as grid upgrades and optimization to handle the extra capacity are inescapable and necessary.

2. Critical Equipment and Supply Chains

Major sectors of the U.S. economy are experiencing shortages and delays with delivery of capital equipment, especially power generation and distribution equipment—and materials. For example, the U.S. lags far behind competitors such as China when it comes to electrical manufacturing. There is only one U.S. producer of grain-oriented electrical steel (GOES), a key component of transformers. Foreign dominance of GOES and other critical transformer components have forced U.S. producers to depend on foreign supply, hurting their ability to compete with more vertically-integrated competitors abroad.[31]

DCC members report broad challenges in procuring critical data center equipment, along with lengthening lead-times for delivery. Shortages are dependent upon general industry demand, leading to demand spikes across all data center operators, including hyperscalers. Some of the critical equipment categories that have experienced delays and fluctuations in delivery timeframes are:

- Mid and high-voltage circuit breakers
- Mid and high-voltage transformers
- Low and mid-voltage switchgear
- Uninterruptible Power Supply (UPS) units
- Chillers and cooling equipment
- Backup generators

While the data center industry works hard to source domestic content, and suppliers are increasingly looking to build more U.S. capacity, it is not always possible to source domestically in the short-term. Supply chain issues for circuit breakers and transformers are ongoing in 2025, and new tariffs previously led to multi-year delays in shipments of switchgear, UPS, and other equipment. We believe the following policy recommendations should be included in the AI Action Plan:

Federal Agency Action:

- **Promote Standardization of Design and Complexity Reduction:** The administration could use DOE's Electricity Advisory Committee (EAC) convening authority to develop common standards for transformer procurement. The default today is that each utility develops different standards and requirements for even commonplace distribution transformers, making standardized sales and manufacturing difficult. Some industry coordination and standard-setting work is already underway through the EAC.

- **Establish a Strategic Transformer Reserve:** Establish a strategic reserve for transformers where the U.S. government can serve as a buyer of last resort if there are manufacturer delays. The reserve's function could be like that of the U.S. Strategic National Stockpile for medical equipment needed to respond to large public health emergencies. As envisioned by the National Infrastructure Advisory Council, the

[31] Green Tape, "How Biden Traded Transformers for Heat Pumps," (February 2025).

mechanism would function like the USDA commodity price supports, except that price would not be the trigger; rather, the support would be activated when plant capacity utilization dips below a particular threshold.[32] The resulting physical reserve could then be stored until production capacity utilization was high and lead times expanded again to unacceptable levels. At that point, the federal government, working with its private sector partners, would determine a suitable reserve drawdown to alleviate shortages.

Congressional Action:

- **Expanding Domestic Capacity through Federal Policies and Funding:** Create incentive structures for critical electrical equipment manufacturing. For example, Congress could pass Senators Moran and Cortez Masto's CIRCUIT Act, which extends a 10% tax credit to transformer manufacturing.[33] This will help domestic transformer production "pencil out" and support the struggling industry as it looks to recapitalize. The executive branch could also repurpose the remaining Defense Production Act (DPA) funding at DOE (~$14M) for transformer manufacturing rather than heat pump manufacturing. This is long overdue, and would support the Trump administration's focus on national security.

- **Expand Domestic Availability of Critical Transformer Components:** Increase availability of GOES and other materials necessary for the construction of transformer components by creating incentives to quickly expand domestic processing capabilities and capacity.

Joint Congressional and Agency Action:

- **Targeted Trade Policy:** Protecting the reliability of the supply chain with careful trade policy that doesn't undermine our ability to source critical equipment is necessary, especially since we are in a race with countries like China for faster digital infrastructure deployment. For example, tariffs on key data center infrastructure components will limit and slow the ability of U.S. companies to make strategic and essential investments to ensure that America remains the world leader in AI. The administration should consider exemptions for key data center components—if only for a defined period.

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[32] The National Infrastructure Advisory Council, "Addressing the Critical Shortage of Power Transformers to Ensure Reliability of the U.S. Grid," (June 2024).

[33] U.S. Congress, Senate, *Credit Incentives for Resilient Critical Utility Infrastructure and Transformers Act (CIRCUIT Act)*, S. 448, 119th Cong., 1st sess., introduced in Senate February 6, 2025.

3. Workforce Development and American Jobs in Data Centers

Investment in domestic data center infrastructure supports growth in domestic jobs, both highly skilled operational jobs in data centers and jobs in the broader economy including construction, electricians, telecommunications, manufacturing, and pipefitters.[34] At the national level, each direct job in the data center industry supports more than six jobs elsewhere in the U.S. economy.[35] Between November 2023 and November 2024, construction spending by the U.S. data center industry increased 43%, compared to a 3% increase in total U.S. construction spending.[36] A recent study by PwC found that the data center industry's total contribution to national labor income was $404 billion in 2023—a 93% increase from 2017.[37] With rapid expansion of AI and the needed data center infrastructure, both trades and highly skilled labor needs will increase even more dramatically.

Worker shortages are commonplace already. A 2023 Data Center Staffing Survey from the Uptime Institute found staffing shortages as high as 48% at mid- and junior-level operations positions, and 41% and 33% in electrical and mechanical jobs, respectively.[38]

As data centers expand from traditional "mature" markets such as Northern Virginia to emerging markets across the country, workforce shortages are amplified by a mismatch between the existing skills in those communities and the skills necessary for operations and trades positions. We believe the following policy recommendations should be included in the AI Action Plan:

<u>Administrative Action:</u>

- **Create a National Technology Hub:** Create a national technology hub for data centers that can train workers, develop common curricula, and coordinate with local and regional academic institutions. The federal government should consider adding a data center operations hub to the "Tech Hubs Program" under the U.S. Department of Commerce.[39]

- **Expand the North American Industry Classification System (NAICS):** The U.S. Office of Management and Budget, the U.S. Census Bureau, and industry should collaborate to expand the classification system with a specific NAICS code for the data center industry.[40] The lack of specific NAICS codes keeps the American workforce in the dark regarding the rapidly growing opportunities for good-paying careers in the data

[34] Fierce Network, "Data center builders are feeling the labor shortage burn," (March 7, 2024).
[35] PwC, "Economic Contributions of Data Centers in the United States."
[36] Associated General Contractors of America, "Construction Outlook Presentation," (January 2025).
[37] PwC, "Economic Contributions of Data Centers in the United States."
[38] The Uptime Institute, "The Uptime Institute Data Center Staffing Survey 2023," (December 2023).
[39] U.S. Economic Development Administration, "Regional Technology and Innovation Hubs (Tech Hubs)."
[40] U.S. Census Bureau, "North American Industry Classification System – NAICS."

center industry. Defining specific NAICS codes for key data center operational careers presents an important opportunity for workforce development.

- **Promote America Works:** DCC supports initiatives to promote skills-based hiring, rather than the traditional approach that uses degrees and certifications. The federal government could partner with business groups and industry to expand skills-based opportunity options in federally sponsored programs.

- **Expand and Strengthen Education Programs:** Build on the success of existing initiatives such as the Northern Virginia Community College program for Data Center Operations,[41] Wenatchee Valley Community College in Washington,[42] and Illinois Community College Board (ICCB) that creates partnerships between the private sector, academia, and the government.[43] Using the existing funding through NSF or the DPA tool of financial assistance, the federal government could provide funding to states, community colleges, or vocational schools, on a regional basis, to expand existing workforce development and training programs or establish such programs for data center operations.

- **Veteran Workforce:** We support programs that train veterans in skills that can be used at data centers. Federal government agencies like the Department of Veterans Affairs can work with industry to establish such programs and provide long-lasting, well-paying jobs to the veteran community.

III. Conclusion

DCC thanks the Trump administration for its leadership in promoting a strong AI ecosystem here in the U.S. Data centers are vital to enabling critical and emerging technologies like AI that are essential to U.S. national security, international competitiveness, and economic prosperity. With the right approach to deploying data center infrastructure, the U.S. can win the AI race safely, securely, and to the benefit of all Americans. We look forward to a continued dialogue and exchanging in-depth ideas about our recommendations on an AI Action Plan.

Please contact Cy McNeill, Director of Federal Energy Policy, at [redacted] if you have questions on the information provided or would like to schedule additional conversations.

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[41] Northern Virginia Community College, "Data Center Operations."

[42] National Science Foundation, "Award Abstract # 2400577, Critical Environments for Data Center Operations," (2024).

[43] DCC and several DCC members serve on the ICCB Advisory Committee. See Illinois Community College Board, "Data Center Curriculum Project Advisory Committee," (April 2024), slide 7.